Synthetic American Option Pricing via Jump-HMM-Driven Heston Implied Volatility

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Abstract

Generating realistic synthetic option prices requires implied volatility (IV) as an input, yet IV itself is derived from observed option prices, creating a circular dependency that limits synthetic data availability for machine-learning and risk-analysis applications. We broke this circularity by developing a three-stage pipeline in which IV emerged as an output of a structural model of equity returns rather than as an exogenous input. The pipeline coupled a Jump Hidden Markov Model that generated multi-asset price paths with realistic stylized facts and cross-asset tail dependence; a modified Heston stochastic variance process that converted those price paths into implied volatility, where the mean-reversion target was driven by the regime state, days to expiration, moneyness, and an aggregate market mood indicator derived endogenously from the fraction of tickers occupying tail states; and a Cox-Ross-Rubinstein binomial tree that priced American options with early exercise using the resulting IV. A central design choice was initializing the variance process at its mean-reversion target for each strike-expiration pair, so that the smile, skew, and term structure of the IV surface emerged automatically without external calibration data. We calibrated the smile and term-structure shape function through a hierarchy of representations that traded off model capacity for sharing across tickers: a parametric baseline established on a single-ticker multi-expiration chain, a globally shared neural surrogate that removed the functional-form bottleneck, and a sector-specific neural surrogate that let within-sector shapes specialize. We applied this hierarchy to a 31-ticker, two-day, six-sector option ladder covering technology, healthcare, financials, energy, retail, and broad-market ETFs, and showed that both added capacity and loosened sharing contributed roughly additive reductions in fit error. The same decomposition supports a natural progression to per-ticker neural surrogates as per-name sample sizes grow. The framework was implemented as an open-source Julia package designed to generate synthetic data for downstream machine-learning and risk-analysis applications.

1 Introduction

Pricing American-style options requires implied volatility (IV) as an input, yet IV itself is an empirical quantity extracted from observed option prices Black and Scholes [1973]. This circularity poses a practical problem for synthetic data generation: existing approaches either take IV from the market and feed it into a pricing model, or generate price paths without a consistent IV surface. Parametric IV models such as SABR Hagan et al. [2002] and the SVI parameterization Gatheral [2004] treat the IV surface as an object to be fitted to market quotes, not generated from first principles; they produce smooth interpolations of observed smiles but cannot generate new IV dynamics for unobserved market conditions. Local volatility models Dupire [1994], Derman and Kani [1994]

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recover a deterministic volatility surface consistent with a single cross-section of option prices, yet the resulting surface is static and cannot produce the stochastic IV fluctuations observed in real markets. Stochastic volatility models such as Heston Heston [1993] generate realistic variance paths, but calibrating them requires observed option prices or IV as input, reintroducing the very circularity that the synthetic generator must avoid. The result is a gap in the literature: no existing framework generates self-consistent IV surfaces as an emergent output of a structural model of equity returns, without requiring observed option data as a calibration target during simulation.

This gap has practical consequences for a growing set of downstream applications. Empirical studies of IV surface dynamics Cont and da Fonseca [2002] have established that implied volatility exhibits rich cross-sectional and temporal structure, including smile curvature, skew, and term-structure effects that interact with market regimes. Reproducing this structure in synthetic data is essential for training machine-learning models that must generalize across market conditions, for backtesting hedging strategies such as deep hedging Buehler et al. [2019], and for conducting scenario-based risk analysis that requires consistent option prices across strikes and expirations. Historical resampling preserves realized distributional features but cannot produce out-of-sample tail scenarios consistent with a structural model of returns; generative approaches based on neural networks have demonstrated promise for equity price paths but do not directly produce self-consistent option price and IV pairs. The scarcity of realistic synthetic option data therefore remains a binding constraint on progress in computational finance.

In this study, we addressed this gap by developing a framework in which IV was an *output* of a structural model of equity returns, not an input. The pipeline coupled a Jump Hidden Markov Model (JumpHMM) Alswaidan and Varner [2025] that produced multi-asset price paths with realistic heavy tails, negligible linear autocorrelation, and persistent volatility clustering Cont [2001], with cross-asset dependence imposed via a Student-t copula Demarta and McNeil [2005]; a modified Heston stochastic variance process Heston [1993] that converted those price paths into IV paths, where the mean-reversion target θ was a function of the HMM regime state, days to expiration (DTE), moneyness, and an aggregate market mood indicator; and a Cox-Ross-Rubinstein (CRR) binomial tree Cox et al. [1979] that converted the pre-computed IV into American option prices with early exercise. The central innovation was a hybrid θ-function that linked the Heston mean-reversion target to the JumpHMM regime state, producing regime-dependent volatility dynamics. An equilibrium initialization $v_0 = \theta(t = 0)$ gave each (ticker, strike, DTE) triple its own initial IV, so the smile, skew, and term structure emerged automatically without external IV data. Within this pipeline, the shape function that controlled the smile and term structure of θ admitted a family of representations, and the modeling choice at each level of the family traded off model capacity for sharing across tickers. We therefore calibrated the shape function through a hierarchy of increasingly expressive forms: a parsimonious five-parameter functional form on single-ticker and cross-ticker semiconductor option chains, a globally shared neural surrogate that replaced the analytic form with a data-driven representation, and a sector-specific neural surrogate that let within-sector shapes specialize. The same decomposition supports a further relaxation to per-ticker networks once per-name sample sizes grow, without changing the downstream pipeline. We demonstrated this progression on a 31-ticker, two-day, six-sector option ladder and showed that the framework produced realistic forward IV paths and American option prices with an endogenous crash-protection premium. The framework is implemented as an open-source Julia package designed to serve as a synthetic option market generator for downstream applications.

2 Background

The framework built on three established models, each contributing a distinct component of the synthetic option generation pipeline: the JumpHMM produced realistic multi-asset price paths, the Heston stochastic variance process provided a mechanism for converting price dynamics into implied volatility, and the CRR binomial tree translated IV into American option prices with early exercise.

2.1 Jump Hidden Markov Model

The Jump Hidden Markov Model (JumpHMM) Alswaidan and Varner [2025] was developed for generating synthetic equity time series that simultaneously preserved three stylized facts of finan-

cial returns: heavy tails, negligible linear autocorrelation, and persistent volatility clustering Cont [2001]. The model discretized continuous excess growth rates into N states via equal-probability quantile bins of a fitted Laplace distribution. The hidden state $S_t \in \{1, \ldots, N\}$ at each timestep governed the distribution of the excess growth rate G_t; each state k emitted observations from a location-scale Student-t distribution: $G_t \mid S_t = k \sim \mu_k + \sigma_k \cdot t_\nu$, where μ_k and σ_k were learned per-state location and scale parameters and ν was the shared degrees-of-freedom parameter. At each timestep, with probability ϵ a Poisson(λ)-distributed jump forced the chain into tail states (bottom or top N_{tail} states), creating sustained dwell periods in extreme-volatility regimes that produced the volatility clustering characteristic of empirical return series. For multi-asset generation, cross-asset dependence was imposed via a Student-t copula Demarta and McNeil [2005] that preserved each asset's marginal distribution while introducing realistic tail dependence; this copula structure ensured that joint tail events, in which multiple assets simultaneously occupied extreme states, occurred more frequently than a Gaussian copula would imply.

2.2 Heston stochastic variance model

The Heston model Heston [1993] specified the instantaneous variance v of the underlying asset as a mean-reverting square-root process:

$$dv = \kappa(\theta - v)\, dt + \sigma_v \sqrt{v}\, dW_v \tag{1}$$

where κ was the mean-reversion speed controlling how quickly the variance returned to its long-run level, θ was the long-run variance target, and σ_v was the vol-of-vol governing the amplitude of stochastic fluctuations in v. In the standard formulation, the asset price and variance processes were coupled through a correlation ρ between their driving Brownian motions, producing the leverage effect in which negative returns coincided with rising volatility. The key property for our purposes was mean-reversion: the variance process was pulled toward θ, so by controlling θ we could control the equilibrium level of implied volatility. In the standard model θ was a constant; our extension made θ time-varying and state-dependent, linking it to the JumpHMM regime state, contract characteristics, and market conditions (Section 3.2).

2.3 CRR binomial tree

The CRR model Cox et al. [1979] priced options on a recombining binomial lattice with up/down factors $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$ and risk-neutral probability $p = (e^{r\Delta t} - d)/(u - d)$. American options were priced by backward induction with an early-exercise check at each node, making the CRR tree a standard tool for contracts where closed-form solutions do not exist. The CRR tree was a natural choice when IV was provided exogenously, as it required only a single volatility input per contract and handled early exercise without the need for Monte Carlo simulation or PDE solvers. Convergence to the continuous-time Black-Scholes price was guaranteed as the number of tree steps increased, and the computational cost scaled linearly in the number of steps, making the tree efficient for pricing large cross-sections of contracts across multiple strikes and expirations.

2.4 Neural representations of the implied volatility surface

Parametric IV models such as SABR Hagan et al. [2002] and SVI Gatheral [2004] expressed the smile and term structure through low-dimensional closed-form functions, chosen to interpolate observed quotes while respecting no-arbitrage constraints. These forms were parsimonious and interpretable but inherited their expressive capacity from the chosen functional family, and a single global parameterization across a broad universe of tickers could not simultaneously accommodate sector-specific skew patterns and smile curvatures. A complementary line of work Horvath et al. [2021], Bayer and Stemper [2019] replaced the analytic form with a neural network that took $(\ln \tau, \ln m)$ or a similar low-dimensional set of contract and model features as input and returned an IV or variance output; the network provided a flexible universal approximator for the shape of the surface while the downstream pricing pipeline remained unchanged. Neural surrogates of this kind have been used both as fast pricing approximators and as calibration targets in deep calibration schemes, and they admit the same multiplicative decomposition between a per-ticker level and a shared shape that the parametric forms use, making them a drop-in replacement for the functional form inside a generative pipeline.

3 Methods

3.1 Pipeline overview

The generator produced synthetic American option prices through a three-stage pipeline:

$$\underbrace{\text{JumpHMM(prices)}}_{\text{Stage 1}} \;\rightarrow\; \{S_t,\, s_t,\, \text{jumps}\} \;\rightarrow\; \underbrace{dv \text{ process}}_{\text{Stage 2}} \;\rightarrow\; \sigma_{\text{imp}} = \sqrt{v_t}$$
$$\rightarrow\; \underbrace{\text{CRR}(S_t,\, \sigma_{\text{imp}},\, K,\, \text{DTE},\, r_f)}_{\text{Stage 3}} \;\rightarrow\; P_{\text{american}} \tag{2}$$

This ordering was dictated by data flow: the JumpHMM produced the price paths and regime-state sequences that the Heston process required as inputs, and the Heston process produced the IV values that the CRR tree required for pricing. Each stage was self-contained, so the pipeline could be run forward without iteration or feedback from downstream stages.

3.2 Modified Heston variance process

We modified the standard Heston process (1) by making the mean-reversion target time-varying:

$$dv = \kappa\big(\theta(t) - v\big)\, dt + \sigma_v \sqrt{v}\, dW_v \tag{3}$$

The mean-reversion target decomposed into a regime-dependent base level, a market mood modulator, and a contract-specific smile and term-structure adjustment:

$$\theta(t) = \theta_{s_t} \cdot \big(1 + \gamma \cdot M_t\big) \cdot \psi(\text{DTE}_t,\; K/S_t) \tag{4}$$

where θ_{s_t} was a learned variance level for HMM state s_t (one value per state, $s_t \in \{1, \ldots, N\}$); $\gamma \geq 0$ was the market mood sensitivity controlling how strongly aggregate stress elevated IV; $M_t \in [0, 1]$ was the aggregate market mood; and $\psi(\text{DTE}, K/S)$ was the smile and term-structure adjustment. The regime-dependent base θ_{s_t} linked the Heston process directly to the JumpHMM state sequence: when the HMM transitioned into a tail state (via a Poisson jump or an extreme Markov transition), θ_{s_t} increased, elevating the IV target. This mechanism produced the leverage effect as an emergent property of the return dynamics rather than as a separate model component: large negative returns drove the HMM into low-numbered tail states with $p_{\text{neg}} \approx 0.52$, which in turn elevated θ.

The function ψ captured the shape of the IV surface with five parameters:

$$\psi(\tau,\; m) = \exp\Big(\beta_1 \ln \tau + \beta_2 \ln m + \beta_3 \ln \tau\, \ln m + \beta_4 (\ln m)^2 + \beta_5 (\ln \tau)^2\Big) \tag{5}$$

where $\tau = \max(\text{DTE}, 1)$ and $m = K/S$. The parameter β_1 controlled the linear term-structure decay; β_2 controlled the skew (negative values produced put skew, with OTM puts having higher IV than OTM calls); β_3 captured the DTE-skew interaction (the skew flattened at longer maturities); β_4 controlled the smile curvature; and β_5 controlled the DTE curvature, capturing the U-shaped ATM term structure where short-dated and long-dated IV were both elevated relative to intermediate maturities.

Rather than specifying a separate initial variance v_0, we initialized the variance process at its mean-reversion target:

$$v_0 = \theta(s_0,\; \text{DTE},\; K/S_0,\; M_0) \tag{6}$$

This ensured that each (ticker, strike, DTE) triple started at its own equilibrium implied volatility. The initial IV surface inherited the smile, skew, and term structure encoded in ψ without requiring any external IV data. Different contracts on the same underlying received different v_0 values, and the variance process immediately began mean-reverting around its local $\theta(t)$. The value of this design choice lay in eliminating v_0 as a free parameter: the model did not require a separate calibration of the initial IV surface, as it was determined by construction from θ.

The market mood M_t was defined as the fraction of tickers currently occupying tail HMM states, where N_{tail} denoted the number of extreme states at each end of the state space (inherited from the JumpHMM fit):

$$M_t = \frac{1}{N_{\text{tickers}}} \sum_{i=1}^{N_{\text{tickers}}} \mathbf{1}\Big\{ s_t^{(i)} \leq N_{\text{tail}} \text{ or } s_t^{(i)} > N - N_{\text{tail}} \Big\} \tag{7}$$

This signal was fully endogenous to the JumpHMM copula model; it required no external data (e.g., VIX) and introduced no circularity. When multiple tickers simultaneously occupied tail states, the mood rose, elevating θ and thus IV across all contracts. This mechanism produced correlated IV spikes during market stress events by construction.

The variance process was discretized via Euler-Maruyama with a reflecting boundary at zero and a daily time step $\Delta t = 1/252$:

$$v_{t+1} = \left| v_t + \kappa(\theta_t - v_t)\Delta t + \sigma_v \sqrt{\max(v_t, 0)}\,\sqrt{\Delta t}\, Z_t \right|, \quad Z_t \sim \mathcal{N}(0,1) \tag{8}$$

The reflecting boundary allowed θ to vary freely across HMM states without requiring the Feller condition $2\kappa\theta > \sigma_v^2$ to hold at every timestep Andersen [2008]. This was a practical necessity, as tail states could drive θ to values where the Feller condition would otherwise be violated. American option prices were computed from the resulting IV using a CRR binomial tree with 200 steps per contract, which provided sufficient convergence for the DTE range considered.

3.3 Neural surrogate for ψ

The parametric form of ψ in equation (5) used five shape parameters shared across all contracts on a given ticker. When the model was applied to a broad universe spanning multiple sectors, this global shape became a bottleneck: the semiconductor smile geometry differed systematically from mega-cap ETFs, healthcare names carried idiosyncratic event risk, and energy tickers inherited commodity-driven skew patterns not expressible by a five-term log-polynomial. We therefore replaced the parametric ψ with a neural surrogate Horvath et al. [2021], keeping the same two inputs $(\ln\tau,\, \ln m)$ and the same multiplicative decomposition $\sigma^2_{\text{model}} = \theta_{\text{ticker}} \cdot \psi(\ln\tau, \ln m)$, but allowing the shape to be data-driven:

$$\psi_{\text{NN}}(\ln\tau, \ln m) = \exp\!\big(\text{NN}(\tilde{z}_\tau,\, \tilde{z}_m)\big), \qquad (\tilde{z}_\tau,\, \tilde{z}_m) = \big((\ln\tau - \mu_\tau)/\sigma_\tau,\, (\ln m - \mu_m)/\sigma_m\big) \tag{9}$$

where $(\mu_\tau, \sigma_\tau, \mu_m, \sigma_m)$ were global standardization constants computed over the full training set. The network produced $\ln\psi$ so that $\psi > 0$ by construction, preserving the log-Gaussian interpretation of the smile surface used in the parametric form. Each network was a two-hidden-layer multilayer perceptron with $\tanh$ activations: $2 \to 16 \to 16 \to 1$ (337 parameters) for groups with ≥ 2000 observations and $2 \to 8 \to 8 \to 1$ (105 parameters) for smaller groups, a width chosen to keep the observations-to-parameters ratio between 10 and 50 across all sector fits.

The decomposition defined a natural hierarchy of sharing. At one extreme, a single global ψ_{NN} shared across all tickers assumed that smile geometry was universal and left only the variance level θ_{ticker} to distinguish names. At the other extreme, a per-ticker ψ_{NN} recovered ticker-specific shape but required enough data per name to resolve the surface without overfitting. We adopted a sector-specific intermediate: one $\psi^{(c)}_{\text{NN}}$ per sector c, with all tickers within a sector sharing shape but each ticker retaining its own θ_{ticker}. This choice reflected the empirical observation that within-sector tickers shared dominant risk factors (technology names tracked semiconductor cycles, healthcare names tracked FDA and trial catalysts, financials tracked rate expectations), so pooling shape across a sector provided a favorable bias-variance tradeoff at the current data volume. As per-ticker sample sizes grew, the same architecture supported relaxation to per-ticker networks without changing the downstream pipeline.

Training minimized the mean squared IV error jointly over the network weights and the per-ticker log-variance levels $\{\ln\theta_{\text{ticker}}\}$:

$$\min_{w,\,\{\ln\theta_t\}} \frac{1}{n}\sum_{i=1}^{n}\Big(\sqrt{\theta_{t(i)} \cdot \psi_{\text{NN}}(\ln\tau_i, \ln m_i;\, w)} - \text{IV}_{\text{market},i}\Big)^2 \tag{10}$$

where $t(i)$ denoted the ticker of the ith observation. The log-variance levels were initialized from each ticker's mean squared market IV and then updated jointly with the network weights under a single Adam optimizer, which coupled the level and shape fits and avoided the chicken-and-egg problem of fitting one conditional on the other. The learning rate followed a step schedule (1e-3 $\to$ 5e-4 at epoch 500 $\to$ 2e-4 at 1000 $\to$ 1e-4 at 1500), and training ran up to 2000 epochs with a best-checkpoint patience of 200 epochs on the training loss. Because the 31-ticker two-day corpus was modest, we did not hold out a temporal validation split at this stage; the reported RMSEs are therefore in-sample fits, and external temporal validation using additional capture days is deferred to future work (Section 6).

4 Calibration and Validation

4.1 Parametric baseline on NVDA

Before replacing the parametric form of ψ with a neural surrogate, we established the baseline capacity of the five-parameter analytic form on a well-characterized single-ticker dataset. We collected NVDA call and put chains at seven expirations on the same observation date (April 6, 2026): 0, 2, 4, 7, 14, 25, and 46 DTE, with NVDA trading near \$176.70, yielding 73 observations after filtering to moneyness $K/S \in [0.90, 1.10]$ with positive volume. The parametric ψ of equation (5) was fitted by minimizing mean squared IV error across all seven expirations simultaneously. The calibrated form captured the full ATM term structure, including the U-shape driven by gamma compression at short maturities and uncertainty premia at longer horizons (Figure 1), achieving an RMSE of 1.15% IV across the 2 to 46 DTE range. The quadratic DTE term β_5 was the critical degree of freedom: without it the linear $\beta_1 \ln \tau$ term could not bend upward at long maturities, and the U-shape was unreachable; with it the parametric form reproduced the trough near DTE ≈ 8 at 32.5% IV and the subsequent rise to 41% at 46 DTE that characterized the market data. This 1% RMSE on a single well-sampled ticker established the upper bound of what the parametric form could deliver under favorable conditions. Additional parametric calibrations, including a single-DTE cross-ticker validation on four semiconductor names, the per-parameter interpretation table, and full residual diagnostics by moneyness and DTE, are reported in the supplement (Section A). Whether the same shape transferred across a broader universe, and how much a more flexible representation could improve the fit when the universe was heterogeneous, motivated the calibration on the full 31-ticker ladder below.

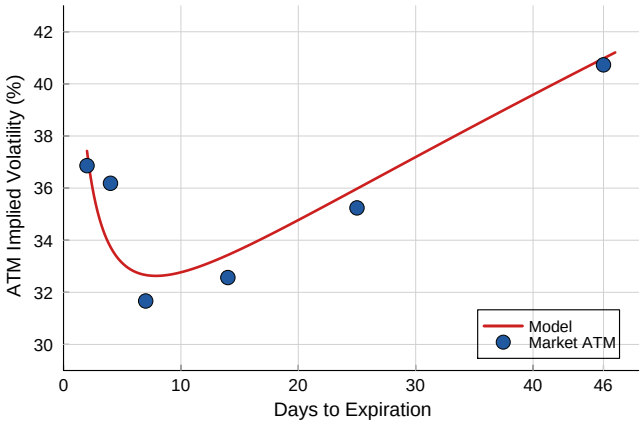


Figure 1: Parametric baseline: ATM implied volatility of NVDA as a function of days to expiration. Market ATM IV (blue markers) exhibited a U-shaped term structure, captured by the five-parameter ψ form (red curve) with a trough near DTE ≈ 8 at 32.5% IV rising to 41% at 46 DTE. The fit achieved an RMSE of 1.15% IV across 73 observations spanning 2 to 46 DTE.

4.2 Sector-specific neural calibration on the 31-ticker ladder

The parametric baseline showed that the five-parameter ψ could reach near-1% RMSE on a single well-sampled ticker, but left open whether the same shape transferred across sectors with heterogeneous risk profiles. To answer this directly, we collected full option ladders for 31 tickers spanning six sectors on two consecutive trading days (2026-04-14 and 2026-04-15), producing 35,961 observations after filtering to moneyness $K/S \in [0.80, 1.20]$, non-zero bids, and IV $\in (0.01, 2.0)$. The sector partition followed conventional GICS-style groupings (Table 1): technology (10 tickers), healthcare (8), financials (4), energy (3), retail (3), and broad-market ETFs (3). This corpus was two orders of magnitude larger than the NVDA multi-expiration dataset and, more importantly, contained systematic cross-sector variation: mega-cap ETFs exhibited shallow smiles with pronounced term structure, technology names carried elevated near-term skew from earnings catalysts, healthcare names showed idiosyncratic bumps around trial and FDA dates, and commodity-linked energy

tickers displayed skew patterns shaped by supply shocks rather than equity crash risk. Against this partition, we quantified the value of relaxing the global-shape assumption by fitting three nested models on the pooled ladder using a common loss (mean squared IV error) and a common level parameterization (one θ_{ticker} per ticker), varying only the representation of ψ: (i) a global parametric ψ with the five β-parameters shared across all tickers; (ii) a global neural ψ_{NN} shared across all tickers, replacing the parametric form with the architecture of Section 3.3; and (iii) a sector-specific neural $\psi_{\text{NN}}^{(c)}$ with one network per sector. This design isolated the contribution of model capacity (parametric $\rightarrow$ NN) from the contribution of sharing granularity (global $\rightarrow$ sector). The three-way comparison revealed that both axes mattered and that their contributions were roughly additive (Table 2; Figure 2a): moving from the global parametric ψ to the global neural ψ reduced overall RMSE from 9.36% to 8.03% IV (a 14.2% relative improvement) by accommodating smile shapes the log-polynomial form could not express, and moving from the global to the sector-specific ψ_{NN} further reduced RMSE to 6.76% (a 27.8% cumulative relative improvement over the parametric baseline) by letting within-sector shapes specialize away from the global average.

Table 1: Sector partition of the 31-ticker ladder. Each ticker appeared on at least one of the two capture days (2026-04-14 and 2026-04-15); per-sector observation counts were pooled across days. ETF observation counts were substantially larger than other sectors because SPY, QQQ, and IWM carried far deeper strike ladders than individual-name chains.

Sector	Observations	Tickers
ETF	17,533	IWM, QQQ, SPY
Technology	7,689	AAPL, AMD, AVGO, GOOG, INTC, META, MSFT, MU, NVDA, QCOM
Healthcare	3,820	ABBV, AMGN, BMY, JNJ, LLY, MRNA, PFE, UNH
Financials	3,496	BAC, GS, JPM, WFC
Retail	1,989	TGT, UPS, WMT
Energy	1,434	CVX, OXY, XOM

Table 2: Three-way overall RMSE comparison on the 31-ticker ladder (35,961 observations). The parametric baseline fit a single global set of five β parameters; the shared NN replaced the parametric ψ with one global network; the sector NN used one network per sector. All three models fitted one θ_{ticker} per ticker jointly with the shape parameters.

Model	Overall RMSE (% IV)	Relative improvement vs. parametric
Parametric (5 β, global)	9.36	(baseline)
Neural ψ (1 network, global)	8.03	14.2%
Neural ψ (6 networks, sector)	6.76	27.8%

The per-sector decomposition exposed where each incremental model capacity helped most (Table 3). The global NN uniformly improved every sector over the parametric baseline, with the largest absolute gains in retail (8.62% $\rightarrow$ 6.94%) and ETF (8.93% $\rightarrow$ 6.89%), where the parametric log-polynomial struggled to fit shallow ETF smiles and idiosyncratic retail patterns simultaneously. Moving to sector-specific networks delivered a further 1 to 2% IV reduction in every sector except technology, which remained the hardest fit (9.16% with sector NN). Inspection of per-ticker residuals (Table 4) traced this to heterogeneity within the sector itself rather than to a failure of the sector NN as a whole: within technology, names such as AMD (5.80%), META (6.75%), and GOOG (6.80%) fit comfortably under the shared Tech surface, while INTC (12.47%), AVGO (12.59%), MU (10.51%), and MSFT (9.76%) pulled the sector RMSE upward. The pattern was consistent with these outlier tickers carrying idiosyncratic smile shapes (elevated wings, asymmetric skew) that a shared Tech ψ could not simultaneously accommodate without degrading the fit of the better-behaved names. The same story repeated in healthcare, where LLY (5.34%) fit well but PFE (13.33%), ABBV (12.18%), and JNJ (11.79%) drove the sector average. The per-ticker θ_{ticker} values recovered a sensible IV level ranking (Figure 2b): high-variance names with concentrated risk (INTC, MU, MRNA, AMD) occupied the top of the sorted order at 60 to 90% baseline IV, large-cap diversified names (AAPL, JPM, WMT) clustered near the middle at 37 to 40%, and the broad-market ETFs (SPY, QQQ, IWM) sat at the low-variance end near 32 to 37%. Within-sector

clustering was visually evident in the sorted bar chart, confirming that sector membership captured a dominant axis of the cross-sectional IV structure.

Table 3: Per-sector RMSE (% IV) across the three calibration strategies on the 31-ticker ladder. The sector NN reduced RMSE in every sector over the shared NN, with the largest absolute gains in sectors where the global shape poorly represented the local geometry.

Sector	Parametric	Shared NN	Sector NN	Obs. count
ETF	8.93	6.89	5.09	17,533
Financials	6.85	6.44	5.54	3,496
Energy	7.66	6.44	5.97	1,434
Retail	8.62	6.94	6.66	1,989
Healthcare	10.78	9.83	8.83	3,820
Technology	11.65	11.23	9.16	7,689

Table 4: Best and worst individual ticker fits under the sector-specific NN calibration, ranked by RMSE across all 31 tickers. Small-sample tickers (PFE, BMY) showed the expected small-sample penalty; the remaining worst fits were concentrated in technology and healthcare where within-sector shape heterogeneity was largest.

Rank	Ticker	Obs. count	RMSE (% IV)	Sector
Best 5				
1	GS	1,755	4.32	Financials
2	WMT	676	4.71	Retail
3	QQQ	6,386	5.06	ETF
3	SPY	7,748	5.06	ETF
5	XOM	498	5.09	Energy
Worst 5				
27	JNJ	408	11.79	Healthcare
28	ABBV	433	12.18	Healthcare
29	INTC	531	12.47	Technology
30	AVGO	594	12.59	Technology
31	PFE	101	13.33	Healthcare

Beyond the aggregate RMSE comparison, the residual and surface diagnostics confirmed that the sector NN behaved sensibly in regions where it mattered most. The residual structure under the sector NN (Figure 3a) showed that the remaining error was concentrated in the wings (moneyness below 0.90 and above 1.10) rather than near the money, consistent with the in-sample fits of the single-ticker baselines and with the known difficulty of pricing deep-OTM contracts where liquidity is thin and quoted IV is noisier. The learned ψ surfaces (Figure 3b) were smooth in both $\ln \tau$ and $\ln m$ and displayed the expected qualitative features, including put skew (ψ rising for moneyness below unity), smile curvature in the wings, and a mild term-structure gradient. The sector-to-sector differences in these surfaces were the direct visual evidence that motivated relaxing the global-shape assumption in the first place: the healthcare and technology surfaces curved more sharply in the near-term wings than the ETF surface, and the energy surface displayed a subtle asymmetry consistent with the commodity-linked risk profile of the sector. This calibration established the sector-specific neural ψ as the current operating point of the framework, but it did not represent the terminal model; the decomposition of Section 3.3 supports per-ticker networks as a direct extension once per-name sample sizes permit. At the 2-day corpus used here, per-ticker sample sizes ranged from 101 (PFE) to 7,748 (SPY), with a median near 600 observations per ticker, sufficient for sector-level fitting but thin for per-ticker networks of the same architecture on most names. The data-rich tickers (SPY, QQQ, IWM, GS, LLY, MSFT) would already support per-ticker fits, and the pattern of within-sector outliers identified above (INTC, AVGO, MU, PFE, ABBV) was precisely the set that per-ticker networks would be expected to help most; additional capture days, reported as future work (Section 6), will support per-ticker ψ_{NN} fits with proper temporal held-out validation across the full universe.

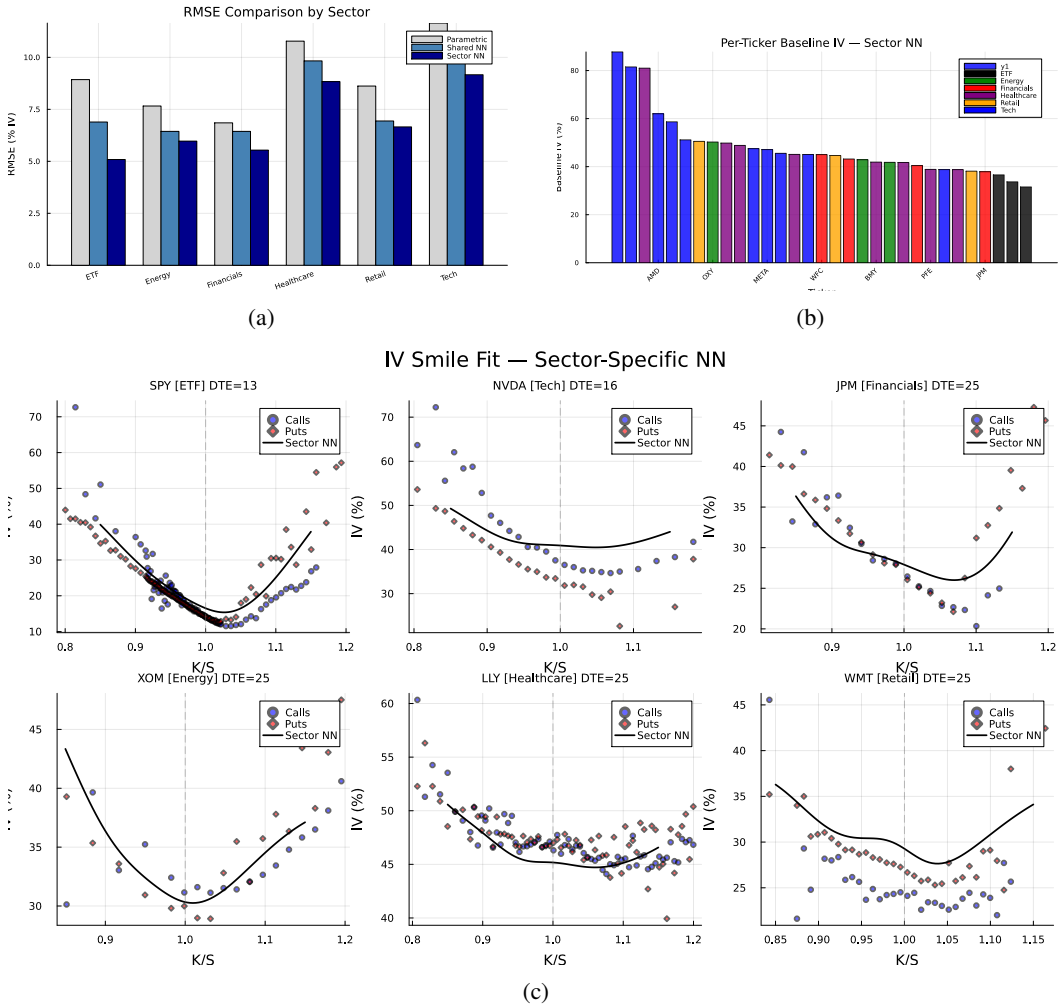
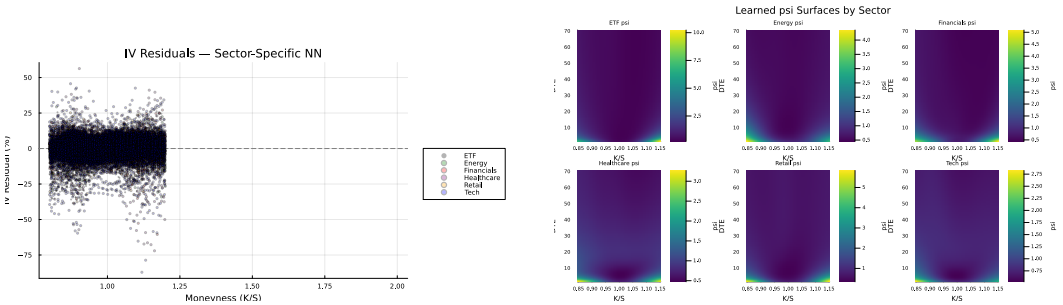


Figure 2: Sector-specific neural calibration on the 31-ticker ladder (35,961 observations, two capture days). **(a)** Per-sector RMSE under the three calibration strategies: parametric ψ (light gray), shared neural ψ (steel blue), and sector-specific neural ψ (dark blue). The sector NN dominated in every sector; the largest gains over the shared NN occurred in ETF and Financials. **(b)** Learned baseline IV ($\sqrt{\theta_{\text{ticker}}}$) for each of the 31 tickers, colored by sector and sorted by level. The ordering recovered the expected high-variance names (MRNA, NVDA) at the top and mega-cap ETFs at the bottom. **(c)** Representative IV smile fits at mid-DTE for one ticker per sector (SPY, NVDA, JPM, XOM, LLY, WMT). Market calls (blue circles) and puts (red diamonds) are overlaid with the sector NN prediction (solid black line). The sector networks captured both the smile curvature and the sector-specific shape asymmetries that a single global ψ could not.

5 Scenario Generation

To demonstrate the framework as a synthetic data generator, we simulated 500 forward paths for NVDA over 80 trading days using the sector-specific neural ψ_{NN} calibrated in Section 4.2 for the technology sector, the corresponding θ_{NVDA} level, and a JumpHMM marginal model with 50 states and tuned jump parameters. The tail states were assigned elevated θ values ($2\times$ baseline for bearish tail states, $1.5\times$ for bullish tail states), and the mood sensitivity was set to $\gamma = 0.5$. The mean IV started near the calibrated ATM level and exhibited gentle term-structure decay as the contract approached expiration, consistent with the positive $\beta_1 = 0.20$ used in the single-DTE calibration (Figure 4). Individual paths showed stochastic fluctuations driven by the vol-of-vol parameter σ_v, with occasional spikes when the HMM transitioned into tail states. The 5th to 95th percentile envelope widened over time, reflecting the accumulation of variance process noise, and the market ATM



(a) (b)

Figure 3: Residual and surface diagnostics for the sector-specific neural calibration. **(a)** IV residuals (model − market) versus moneyness, colored by sector. Residuals clustered around zero in the near-money range ($K/S \in [0.90, 1.10]$) and dispersed in the wings, consistent with reduced liquidity and noisier quoted IV for deep-OTM contracts. **(b)** Learned ψ surfaces over $(K/S, \text{DTE})$ for each of the six sectors, plotted on a shared color scale. The surfaces were smooth in both inputs and displayed sector-specific differences in skew steepness and smile curvature that motivated the sector-level relaxation of the shared-shape assumption.

IV of 44.6% sat within the model's distributional range and confirmed that the calibrated parameters produced IV dynamics centered on the observed level.

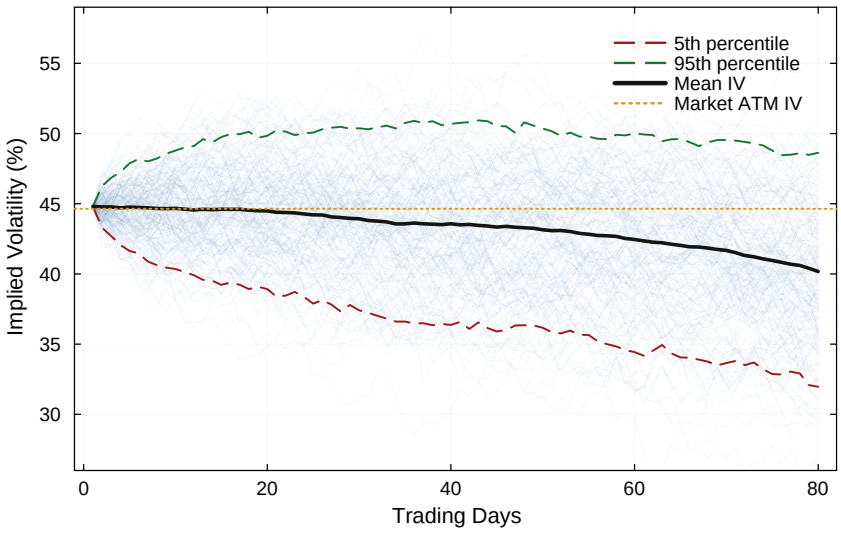


Figure 4: Forward IV paths for NVDA (500 scenarios, DTE = 80). The solid black line shows the mean IV; dashed red and green lines mark the 5th and 95th percentiles; the dotted orange line indicates the market ATM IV (44.6%). Semi-transparent blue paths show individual scenario trajectories with stochastic fluctuations and occasional tail-state-driven spikes.

To translate these IV dynamics into contract valuations, we computed American option prices across five strike levels at $t = 5$ days. The price distributions exhibited realistic properties (Table 5): OTM puts (strike $= 0.90S$) had a mean price of \$12.59 with a 90% confidence interval of [\$5.6, \$21.1]; ATM calls were approximately $2\times$ more expensive than 10%-OTM calls; and the confidence intervals widened for contracts farther from the money. The market mood mechanism produced a natural partition of scenarios into calm ($M_t = 0$) and stressed ($M_t > 0$) paths. In multi-asset simulations with five tickers, stressed scenarios (where multiple tickers simultaneously occupied tail HMM states) produced systematically higher option prices. OTM puts were $1.83\times$ more expensive in stressed scenarios relative to calm scenarios (strike $= 0.95S$: \$6.42 compared with \$3.51), while deep OTM weekly puts showed a $4.33\times$ stress premium. This asymmetry, with puts repricing

more sharply than calls under stress, was consistent with the crash-protection premium observed in equity options markets Bates [2000] and emerged naturally from the JumpHMM's negative jump bias ($p_{\text{neg}} = 0.52$) propagating through the θ-function.

Table 5: American option price distributions from 500 forward scenarios (NVDA, evaluated at $t = 5$). Strike levels are expressed relative to the spot price $S = \$207$.

Type	Strike	DTE	Mean ($)	Std ($)	90% CI ($)
Put	186 ($0.90S$)	80	12.59	4.99	[5.6, 21.1]
Put	197 ($0.95S$)	80	16.69	5.56	[8.5, 26.2]
Call	207 (ATM)	80	25.59	6.64	[15.5, 37.0]
Call	217 ($1.05S$)	80	20.94	6.29	[11.4, 31.7]
Call	228 ($1.10S$)	80	16.72	5.81	[8.1, 26.6]

6 Discussion and Future Work

The calibration and scenario results validated the core claim of this study: a modified Heston process with a state-dependent θ-function produced realistic IV surfaces from structural return dynamics alone, and the same decomposition of θ into a per-ticker level and a shape function ψ accommodated a hierarchy of representations spanning a parsimonious analytic form, a globally shared neural surrogate, and a sector-specific neural surrogate. The single-DTE parametric smile captured the essential features of the NVDA IV smile, and the smile shape generalized across semiconductor names with only a scalar level adjustment; the multi-expiration calibration extended this to the full NVDA IV surface with the five-parameter ψ function and the quadratic DTE term β_5 that captured the U-shaped ATM term structure. The equilibrium initialization $v_0 = \theta(t = 0)$ eliminated v_0 as a free parameter and produced the smile, skew, and term structure by construction; this was not merely a computational convenience but a structural constraint that ensured the initial IV surface was fully determined by the θ-function, consistent with the view that IV surfaces reflect the conditional distribution of returns under the risk-neutral measure Cont and da Fonseca [2002]. The ψ-function played an analogous role to the SVI parameterization Gatheral [2004] in that it provided a parsimonious representation of the smile and term structure, but with a critical difference: the parameters entered a generative model rather than a static interpolation, so the calibrated surface could evolve dynamically as the underlying HMM state changed.

Scaling the calibration to the 31-ticker two-day ladder revealed the limits of the parametric form as a universal shape function and motivated the hierarchy of neural representations. A single global ψ fit across the six sectors achieved an overall RMSE of 9.36% IV, dominated by systematic misfits in sectors whose smile geometry departed from the semiconductor baseline (healthcare at 10.78%, technology at 11.65%). Replacing the parametric form with a globally shared neural ψ reduced the overall RMSE to 8.03% without changing the sharing granularity, confirming that part of the gap was a functional-form bottleneck in the log-polynomial representation. Loosening the sharing to one network per sector reduced the overall RMSE further to 6.76%, with the sector NN strictly dominating the shared NN in every sector, which established that both axes (capacity and granularity) contributed independently. The remaining error was concentrated in two structural sources: within-sector heterogeneity in technology and healthcare, where specific tickers (INTC, AVGO, MU, PFE, ABBV) carried idiosyncratic smile shapes not expressible by a shared sector-level ψ, and small-sample effects for tickers with fewer than 200 observations (PFE at 101 observations). Both sources are addressable through the same decomposition: the per-ticker roadmap relaxes shape sharing to the ticker level for the data-rich names, and additional capture days directly reduce small-sample variance for the thin names.

Several limitations constrained the scope of the current conclusions and pointed directly to the natural extensions that followed from them. The 31-ticker calibration reported in-sample RMSEs because a single two-day snapshot did not support a temporal held-out split; early stopping was used as a convergence criterion rather than as a true held-out-validation mechanism. Planned capture days will support a proper train-on-two-days, test-on-third-day evaluation that separates fit from generalization, and will simultaneously enable per-ticker neural ψ fits for the data-rich names where the 2-day corpus is already sufficient. The cross-sector partition used conventional GICS-style group-

ings; whether alternative groupings (mega-cap versus semiconductor within technology; pharma versus biotech within healthcare) improve the sector-NN fit without fragmenting the corpus is an empirical question that additional data will clarify. The forward scenario analysis (Section 5) confirmed that the calibrated parameters produced IV paths centered on observed market levels with realistic stochastic fluctuations, but the validation was limited to a single ticker and a single market regime; broader validation across sectors and volatility environments, and a re-run of the scenarios with the sector-NN ψ for each represented sector, would strengthen the generalizability claim. Beyond these specific limitations, the primary intended use of this framework was generating realistic synthetic training data for machine-learning models in finance, with applications ranging from training option price predictors and learning delta-hedging policies via reinforcement learning Buehler et al. [2019] to augmenting scarce real option data for transfer learning. The framework naturally supported computation of Greeks (Δ, Γ, $\mathcal{V}$, Θ) via finite differences on the CRR tree, and including these in the synthetic dataset would support hedging applications. Finally, the current mood mechanism created correlated IV dynamics through a shared stress signal; a richer model could introduce ticker-pair IV correlations through the copula structure, producing a full cross-asset IV surface that respected observed correlation patterns between individual-name and index volatility.

7 Conclusion

In this study, we developed a framework for generating synthetic American option prices with self-consistent implied volatility dynamics. The framework linked a Jump Hidden Markov Model to a modified Heston stochastic variance process via a state-dependent mean-reversion target $\theta(t)$, and priced the resulting IV through a CRR binomial tree with early exercise. The equilibrium initialization $v_0 = \theta(t{=}0)$ eliminated the initial IV surface as a free parameter, and the endogenous market mood signal produced correlated IV spikes during stress events without external data.

We calibrated the smile and term-structure shape function ψ through a hierarchy of representations. A five-parameter parametric form fitted on NVDA options first demonstrated that the smile shape generalized across AMD, MU, and INTC with cross-ticker RMSE of 1 to 5% IV using only a scalar level adjustment per ticker, and a multi-expiration fit across seven expirations revealed a U-shaped ATM term structure captured by the quadratic DTE extension. Scaling to a 31-ticker, two-day, six-sector option ladder exposed the limits of the parametric form as a universal shape function, and we demonstrated that replacing it with a globally shared neural surrogate and then with sector-specific neural surrogates reduced the overall fit error by 14% and 28% respectively, with independent contributions from added model capacity and loosened sharing granularity. The same decomposition supports a natural progression to per-ticker networks once per-name sample sizes grow, and the forward scenario analysis confirmed that the generated IV dynamics exhibited realistic stochastic fluctuations and a crash-protection premium in OTM puts under stress (1.8 to 4.3$\times$ calm-to-stress price ratios). The framework was implemented as an open-source Julia package and is available at `https://github.com/varnerlab/heston-implied-volatility-model`.

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A Parametric baseline: additional calibration and diagnostics

The main text reports a single ATM term-structure fit to establish the parametric ψ as a baseline (Section 4.1). This supplement reports the two earlier-stage parametric calibrations that preceded the neural surrogate: a single-DTE fit on four semiconductor tickers that demonstrated cross-ticker transfer of the smile shape, and the full multi-expiration NVDA calibration from which the main-text figure was drawn, including the parameter values, residual diagnostics, and IV surface reconstruction.

A.1 Single-DTE four-ticker semiconductor calibration

The empirical starting point for the parametric form was a set of option chains for four semiconductor stocks: NVDA, AMD, MU, and INTC. All chains were single-date snapshots sourced from barchart.com (October to November 2025) with DTEs ranging from 41 to 80 days, filtered to moneyness $K/S \in [0.70, 1.30]$ and IV $< 200\%$ to exclude illiquid deep ITM/OTM contracts with unreliable IV (Table 6). Because each chain was a single-date snapshot, all observations shared one HMM state s_0 and market mood M_0, so the model IV at a given strike reduced to $\sigma_{\text{model}}(K) = \sqrt{\theta_{\text{base}} \cdot \psi(\text{DTE},\ K/S)}$, where $\theta_{\text{base}} = \theta_{s_0} \cdot (1 + \gamma M_0)$ absorbed the state and mood into a single scalar. We fixed $\beta_1 = 0.20$ (term-structure slope) because single-DTE snapshots could not identify it; the remaining parameters $(\theta_{\text{base}}, \beta_2, \beta_3, \beta_4)$ were fitted by minimizing the mean squared IV error on NVDA using Nelder-Mead optimization. We chose NVDA as the calibration ticker because it had the richest chain (142 contracts after filtering).

The calibrated parameters (Table 7) produced the expected smile features: the negative β_2 generated put skew, the positive β_4 generated smile curvature, and the interaction term β_3 captured how the skew flattened at longer maturities. The calibration achieved an RMSE of 1.16% IV on NVDA. To determine whether the smile shape generalized across tickers, we applied the $(\beta_1, \beta_2, \beta_3, \beta_4)$ calibrated on NVDA to AMD, MU, and INTC, re-optimizing only the scalar θ_{base} per ticker. The NVDA fit confirmed that four parameters sufficed to capture the smile shape within the tradeable moneyness range (Figure 5a). The cross-ticker validation revealed that the shared smile shape transferred well to AMD (2.34%) and reasonably to MU (3.99%) and INTC (4.65%; Table 8; Figure 5b). The higher

Table 6: Option chain summary for the single-DTE semiconductor dataset. Contract counts are reported after filtering to the tradeable moneyness range.

Ticker	Spot ($)	DTE	ATM IV (%)	Contracts
NVDA	207.04	80	44.6	142
AMD	225.78	58	59.0	36
MU	237.02	41	72.4	34
INTC	37.78	65	53.6	46

RMSE for MU and INTC reflected differences in skew steepness across these names; MU exhibited a steeper put skew than NVDA, while INTC had a flatter OTM call wing, suggesting that per-ticker β calibration would further reduce errors when multi-strike data was available. The residual analysis confirmed that the errors were concentrated in the wings ($K/S < 0.85$ and $K/S > 1.15$), with near-ATM residuals below 1% IV for all tickers (Figure 5c). These results established that the ψ-function captured the dominant smile features with a small number of parameters, and motivated the multi-expiration extension reported below.

Table 7: Calibrated smile parameters from the single-DTE NVDA fit (DTE = 80).

Parameter	Interpretation	Value
β_1	Term-structure decay (fixed)	0.20
β_2	Skew (asymmetry)	-2.07
β_3	DTE $\times$ skew interaction	0.34
β_4	Smile curvature	1.43
θ_{base}	Variance level (NVDA)	0.084

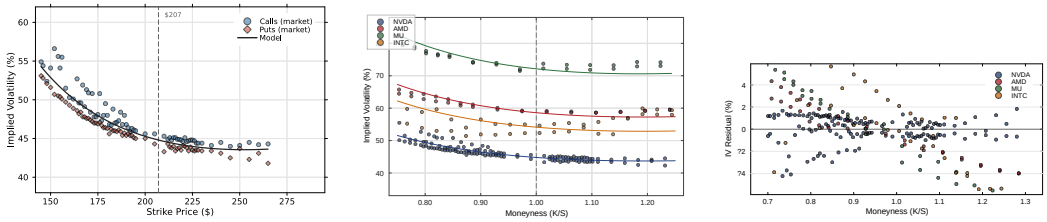
Table 8: Cross-ticker RMSE (% IV) from the single-DTE calibration. The smile shape $(\beta_2, \beta_3, \beta_4)$ was shared from NVDA; only θ_{base} was re-fitted per ticker.

Ticker	θ_{base} (IV level)	RMSE (% IV)
NVDA (calibration)	28.9%	1.16
AMD	38.7%	2.34
MU	49.0%	3.99
INTC	35.9%	4.65

A.2 Multi-expiration NVDA calibration

Having validated the smile shape across tickers, we addressed the term-structure limitation by collecting NVDA call and put chains at seven expirations on the same observation date (April 6, 2026): 0, 2, 4, 7, 14, 25, and 46 DTE, with NVDA trading near $176.70. The cross-sectional snapshot provided the multi-expiration data needed to identify β_1 directly and to resolve the full term-structure geometry, yielding 73 observations after filtering to moneyness $K/S \in [0.90, 1.10]$ with positive volume. The six parameters $(\sigma_{\text{base}}, \beta_1, \ldots, \beta_5)$ of equation (5) were fitted by minimizing mean squared IV error across all seven expirations simultaneously. The single ATM term-structure figure in the main text (Figure 1) was drawn from this fit.

The calibrated coefficients (Table 9) combined a negative β_1 with a positive β_5 to produce the U-shaped ATM curve: at short maturities the linear term dominated (IV decayed rapidly from the 0 DTE spike), while at longer maturities the quadratic term bent the curve upward. The trough occurred near DTE ≈ 8, where IV reached 32.5%, a 1.38$\times$ ratio between the 0 DTE peak and the trough. The skew parameters ($\beta_2 = -8.45$, $\beta_3 = 1.62$) confirmed that the volatility skew steepened at shorter maturities and flattened as DTE increased, consistent with the concentration of crash risk premia in near-term options Bates [2000] (Figure 6a). The overall fit achieved an RMSE of 2.31% IV across all seven expirations and an RMSE of 1.15% IV across the 2 to 46 DTE range reported in the main text. The fit was tightest at intermediate and long maturities (7 to 46 DTE: RMSE $< 1.3\%$), with the largest residuals at 0 DTE (RMSE = 6.57%) where the gamma-compressed



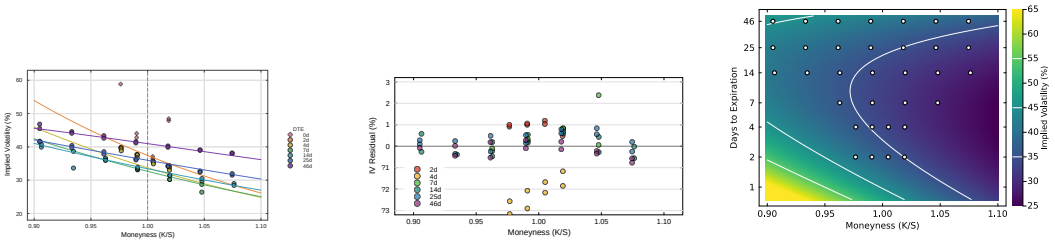
(a) (b) (c)

Figure 5: Single-DTE parametric calibration and cross-ticker validation. **(a)** NVDA implied volatility smile (DTE = 80): market calls (blue circles) and puts (coral diamonds) vs. the fitted model curve (solid line); the dashed vertical line marks the spot price ($207). **(b)** IV smile fit across all four semiconductor tickers as a function of moneyness (K/S). The smile shape $(\beta_2, \beta_3, \beta_4)$ was shared from NVDA; only the scalar level θ_{base} was re-fitted per ticker. **(c)** IV residuals (model − market) by moneyness across all four tickers. Errors were concentrated in the wings ($K/S < 0.85$ and $K/S > 1.15$), with near-ATM residuals below 1% IV for all tickers.

smile shape deviated from the log-moneyness functional form (Figure 6b). The 0 DTE level bias was eliminated (-0.1%), confirming that the U-shaped term structure was captured correctly even though the extreme smile curvature at expiration remained a source of scatter (Figure 6c).

Table 9: Calibrated ψ parameters from the multi-expiration NVDA dataset.

Parameter	Interpretation	Value
σ_{base}	Base ATM IV level	45.0%
β_1	DTE decay	-0.622
β_2	Skew (asymmetry)	-8.448
β_3	DTE × skew	1.617
β_4	Smile curvature	-1.242
β_5	DTE curvature	0.150



(a) (b) (c)

Figure 6: Multi-expiration parametric calibration diagnostics (NVDA, 2 to 46 DTE). **(a)** IV smile curves at each of six expirations (2, 4, 7, 14, 25, 46 DTE). Market observations (markers) and model predictions (lines) are colored by DTE; 0 DTE data (diamonds) were excluded from the fit due to gamma compression. The skew flattened with increasing maturity, consistent with $\beta_3 > 0$. **(b)** IV residuals (model − market) by moneyness, colored by DTE. Most residuals fell within ±2% IV, with no systematic bias across the moneyness range. **(c)** Model-predicted IV surface as a function of moneyness and DTE (log-scaled). White contour lines mark 35%, 45%, and 55% IV levels; white circles indicate market observations used for calibration. The overall RMSE was 1.15% IV across the 2 to 46 DTE range.